PowerAware – Low-Power Domain Design with UPF

- Project Presentation
- Goal: Design a low-power digital subsystem, implement power intent using UPF, and validate shutdown, wake-up, isolation and retention behavior through simulation and checks.

1. Project Overview

• Design a low-power digital subsystem using modern power-intent methodology.

• Partition the design into always-on, switchable and retention regions.

• Describe power intent with Unified Power Format (UPF).

• Validate power-state transitions, isolation and retention through simulation and checks.

• Summarize power/performance trade-offs and final project results.

2. Key Objectives & Technical Architecture

- Power Domains – Always-on, switchable and retention regions.
- UPF Authoring – Power switches, isolation and retention strategies.
- Verification – Low-power rule checks and simulation scenarios.
- Optimization – Compare baseline and low-power estimates.

3. Phase 1 – Power Architecture Planning

- Partition the design into logical power domains.
- Define an always-on domain for control and power-management logic.
- Define a switchable domain for logic that can be powered down to reduce leakage.
- Identify retention requirements for state that must survive a power-down event.
- Plan power states and transition rules before UPF implementation.

4. Phase 2 – UPF Implementation

- Create UPF power domains and associate RTL instances with each domain.
- Add power switches to control the switchable domain.
- Add isolation cells to protect always-on logic when the switchable domain is OFF.
- Define retention strategies for state-holding elements.
- Connect power states and transition rules to describe legal operating modes.

5. Phase 3 – Verification & Simulation

- Run low-power rule checks to validate UPF intent and domain connectivity.
- Run simulation scenarios covering normal operation, shutdown and wake-up.
- Verify isolation behavior when the switchable domain is powered down.
- Verify retention behavior across power-down and wake-up.
- Use waveform results to confirm correct power-state transitions.

6. Final Output & Results

• Functional result – RTL + UPF simulation produces the expected functional behavior.

• Low-power result – Power-down and wake-up sequences are represented in the simulation flow.

• Verification result – Waveform evidence and low-power checks support intended power-state behavior.

• Final artifacts – Source code, simulation/VCD files, final outputs and documentation are organized in the project folder.

7. Phase 4 – Power Optimization Summary

- Compare the baseline design with the low-power implementation.
- Evaluate the effect of power switches, isolation and retention cells.
- Report area and timing impact introduced by low-power cells.
- Summarize the power/performance trade-off and identify the benefit of power-domain shutdown.
- Use measured/simulated project results rather than assuming ideal savings.

8. Deliverables

- 1. UPF files and power-domain documentation.
- 2. Low-power verification logs and simulation evidence.
- 3. Power/performance trade-off report.
- 4. RTL/source code and supporting project files.
- 5. Final presentation and project handoff package.
- Project organization: Documentation/, source_code/, Simulation/, Final/, Github/, presentation/

9. Project Timeline & Deadlines

- Week 1: Domain partition and power strategy.
- Week 2: UPF implementation.
- Week 3: Low-power verification.
- Week 4: Analysis and final handoff.

10. Conclusion

- PowerAware demonstrates a practical low-power design flow using UPF.
- Power domains define when logic is powered and controlled.
- Isolation and retention preserve correct behavior across power-state changes.
- Simulation and low-power checks provide verification evidence.
- The final analysis captures power benefits together with area/timing trade-offs.